

Predicting and Preventing Revenue Compression

Proactive Customer Retention through Machine Learning



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The Economics of Retention in Telecom

Market Reality

5X to 25X !!

According to research by Bain & Company, it costs 5 to 25 times more to acquire a new customer than to retain an existing one. Furthermore, increasing customer retention rates 5% can increase profits by 25% to 95%

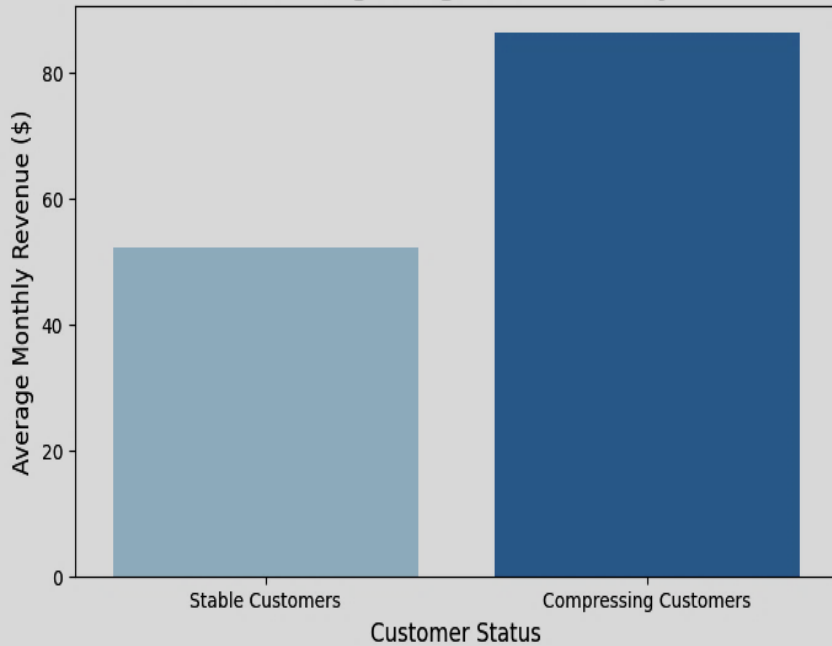


The Hidden Threat of "Revenue Compression"

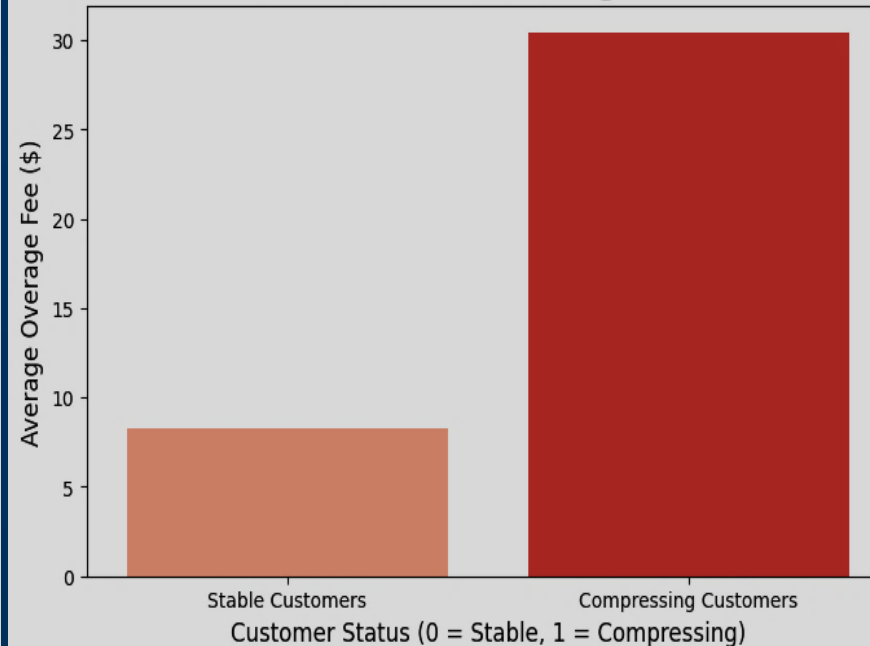
- 1:** Most telecom operators strictly monitor "hard churn" (when a customer completely leaves the network). However, our data suggests this client is suffering from a massive, quieter leak: Revenue Compression.
- 2:** High-value, premium customers are staying on the network but quietly downgrading their expensive plans to cheaper, lower-tier data packages.
- 3:** Because replacing the lost profit margins of a premium user requires expensive new campaigns, our objective is to use machine learning to predict and stop these plan downgrades *before* they happen.

Data Insights: Identifying the Root Cause

Who is downgrading? (Base Monthly Bill)



The "Bill Shock": Overage Fees



Conclusion:

Premium customers are reacting to sudden "bill shock" by downgrading their base plans to avoid future penalties. We must build a predictive AI model to catch these overage spikes before the customer's bill is finalized.

High-Value Volatility: Raw data analysis indicates that plan downgrades are not a budget-tier problem. The customers at the highest risk of compression are premium users (average base bill of ~\$90).

The "Bill Shock" Trigger: A massive anomaly exists in the overage data. Customers who compress their plans experience a sudden **\$35 penalty fee spike** immediately prior to downgrading.



Translating the Business Problem into an ML Task

Proactive Intervention

We must shift from reactive customer service to proactive retention. The objective is to deploy an algorithm capable of analyzing current-month billing anomalies to flag "Bill Shock" victims *before* they initiate a plan downgrade



Binary Classification

We established a binary target variable
1 = Compressing | 0 = Stable.
We removed post-event data columns from the training set to prevent "target leakage," ensuring the model only trains on predictive, historical clues rather than the answers.

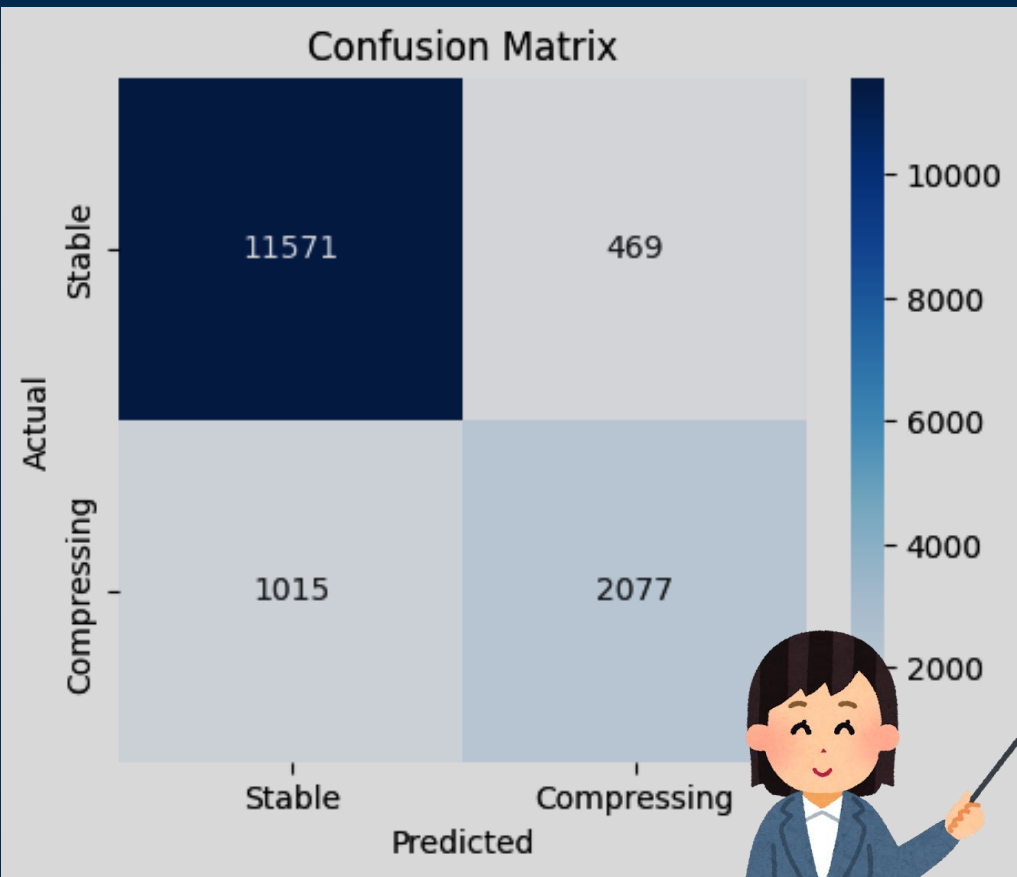


Feature Engineering

To prepare the raw telecom dataset for machine learning, we applied Categorical Encoding (translating text like device types into numerical formats) and utilized median imputation to handle missing data without distorting the underlying behavioral patterns.



Model Performance: Validating the Proof of Concept



Algorithm Selection: XGBoost Classifier:

We utilized the XGBoost (Extreme Gradient Boosting) algorithm, selected for its state-of-the-art ability to map complex, non-linear customer behaviors within structured telecom datasets.

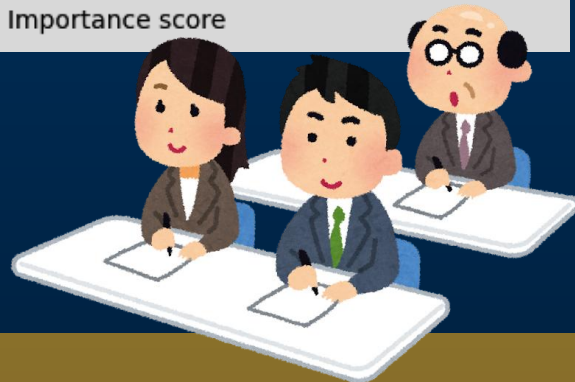
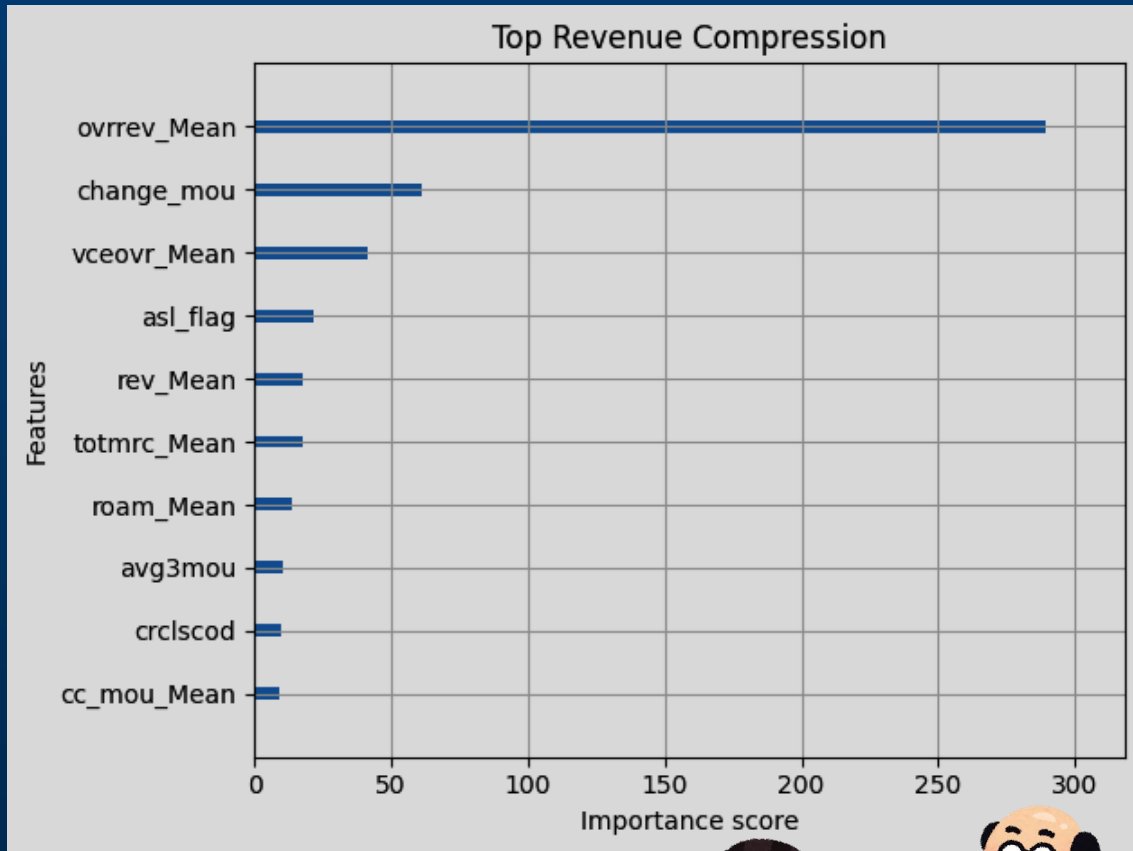
Primary Metrics: 90.2% Accuracy & 0.7368 F1Score:

While the model achieved high overall accuracy, we optimized for the **F1 Score**. In an imbalanced dataset where most customers are stable, the F1 score ensures we are highly precise in catching the specific minority of users who are actually at risk of a revenue leak.

The "Trust" Factor (Precision):

As shown in the matrix, when the model raises an alarm, it is correct **81.5% of the time**. This high precision prevents the Customer Success team from wasting resources on "false alarms" and ensures we only offer discounts to customers who truly need them.

Decoding the Model: The Primary Driver



Primary Driver: The XGBoost algorithm allows us to see exactly how it makes decisions. The model overwhelmingly identified `ovrrrev_Mean` (Surprise Overage Revenue) as the **#1** predictor of plan downgrades—making it four times more predictive than any other behavioral change.

Validating the Hypothesis: This mathematical output perfectly confirms our initial EDA hypothesis. The model proves that "Bill Shock" is not just a correlation; it is the root behavioral cause of our revenue compression.

Actionable Intelligence: Because the model tells us exactly *why* these 2,077 customers are at risk, we don't just have a list of names—we have the exact leverage needed to design a highly targeted retention campaign.

Bill Shock Mitigation Plan

Detection (Flag)

Proposed Integration: Deploy the XGBoost predictive model to run alongside the monthly billing cycle. The system is designed to scan premium accounts prior to statement finalization, specifically isolating accounts that have triggered sudden overage penalties.

Targeted Routing (Reach)

Operational Workflow: Flagged accounts (such as the 2,077 at-risk users identified in this test group) are automatically routed to the Customer Success team as high-priority leads, enabling intervention before the customer views their penalized bill.

Strategic Intervention(Retain)

The Retention Offer: Representatives proactively contact the user to offer a one-time "Fee Waiver." The penalty is forgiven on the condition that the user maintains their current premium plan, neutralizing the frustration of the overage charge.

The Result: We eliminate the "Bill Shock" trigger entirely. The customer feels valued rather than penalized, brand loyalty increases, and the company secures the long-term recurring revenue of the premium plan.



Quantified Financial Impact & ROI

Protected Revenue:
\$186,680 / mo

Gross monthly recurring revenue secured from the 2,077 at-risk premium accounts identified by the model.

Campaign Investment:
- \$84,848 / mo

The direct cost of waiving the surprise overage fees for these specific flagged accounts.

Net Monthly Impact:
\$101,832 / mo

Net retained monthly profit margin directly attributed to this proactive machine learning intervention

Financial Assumptions & Model Parameters :

Assumption 1: Calculations are based on the test data subset. The model successfully isolated 2,077 true positive cases of imminent downgrades.

Assumption 2: Assumes an average premium base plan value of ~\$90/month and an average overage penalty waiver cost of ~\$40 per flagged account.

Assumption 3: Assumes a 100% acceptance rate of the fee waiver offer. Even with a conservative 50% acceptance rate, the model still yields a net positive return of over \$50,000 monthly, far outweighing new customer acquisition costs.



Strategic Implementation & Risk Mitigation

Model Limitations (False Positives)

The Risk: While highly precise, the model has an 18.5% false-positive rate. This means a small portion of flagged accounts are "stable" customers who were not actually planning to downgrade, resulting in unnecessary fee waivers.

The Mitigation: These false positives are still our highest-value premium customers. Providing them with a proactive, surprise fee waiver acts as an unexpected "surprise and delight" loyalty investment, fundamentally reducing their long-term churn risk. The net ROI remains overwhelmingly positive.

Recommended Next Steps

- 1. IT Integration (Month 1):** Deploy the XGBoost algorithm to the central data warehouse to run automated batch scoring 72 hours prior to billing finalization.
- 2. Controlled Pilot (Month 2):** Launch the retention campaign to 50% of the flagged at-risk accounts. Compare their retention rates against the remaining 50% (control group) to validate the exact real-world financial return.
- 3. Full Scale (Month 3):** Fully automate the data pipeline, routing flagged accounts directly into the Customer Success CRM dashboards for daily agent intervention.



References & Documentation

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